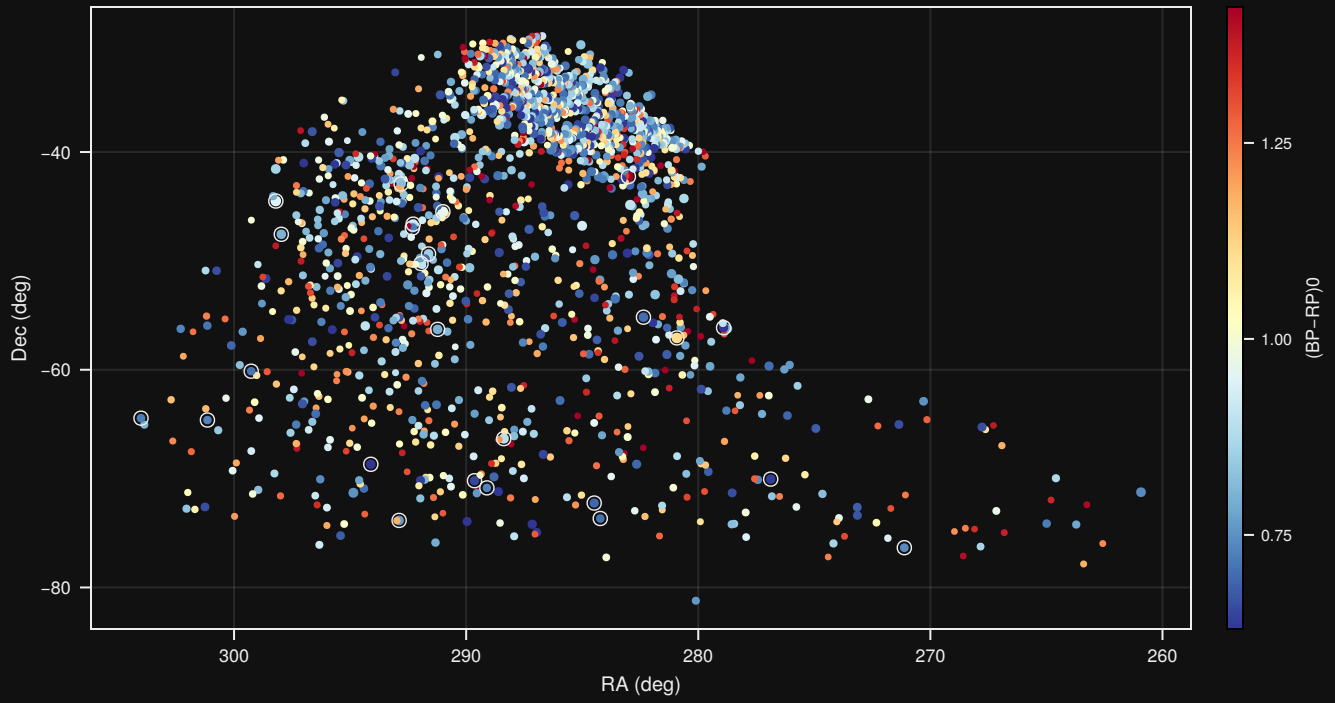
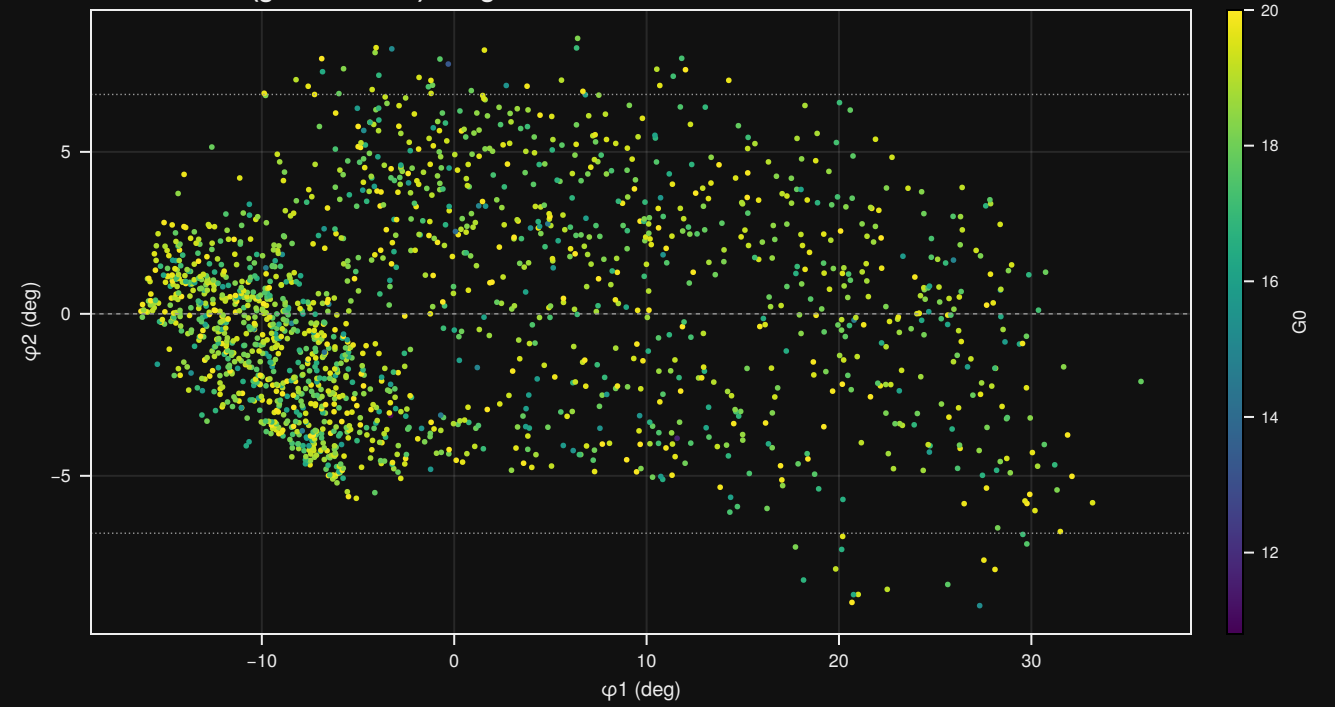
Gaia-9

MEMBERSHIP	
Stars in catalog	1734
with Gaia GSP-Phot	923 (53%)
with radial velocity	25
Stream id (sID)	55
GEOMETRY · great-circle fit	
Centroid RA, Dec	287.4, -40.7 deg
Centroid l, b	351.5, -19.9 deg
Pole RA, Dec	16.2, 1.2 deg
Length (φ1, 2–98%)	43.3 deg
Width σ(φ2), robust	3.39 deg
KINEMATICS	
pmRA* med ± σ	-9.30 ± 3.30 mas/yr
pmDE med ± σ	-23.27 ± 4.26 mas/yr
μφ1* med ± σ	23.39 ± 4.06 mas/yr
μφ2 med ± σ	-10.50 ± 3.40 mas/yr
Parallax (cat.) med	0.319 ± 0.232 mas
Parallax (wtd, ZP-corr.)	0.343 ± 0.010 mas
Distance	d = 2.9 ± 0.1 kpc
Physical length	2.21 kpc
RV median ± σ(MAD)	7.4 ± 145.8 km/s
RV gradient dV/dφ1	8.65 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	10.8–20.0 (18.6)
(BP−RP)0 median	0.93
GSP-Phot [M/H] (n=923)	-1.87 ± 0.55
GSP-Phot Teff median	5400 K
RUWE > 1.4	0.8 %
PM clusters (largest)	120 (40 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	1734 | 233
n_RV catalog | paper	25 | 25
<[Fe/H]> paper	-2.21
Stellar mass (paper)	950 Msun
Pole offset vs paper	29.4 deg
LABEL CHECK	n differs from paper!

Sky map · catalog members (n = 1734) · size ∝ brightness · ring = has RV

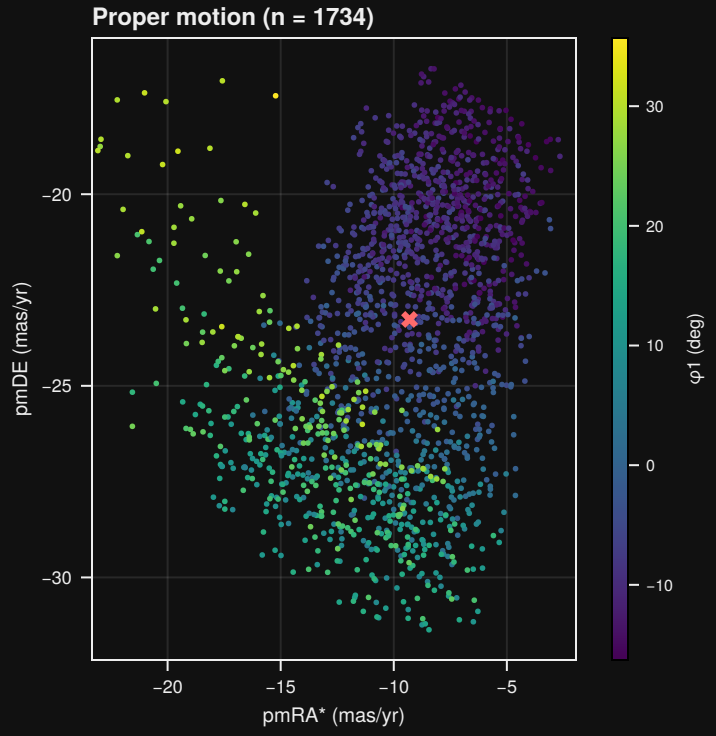


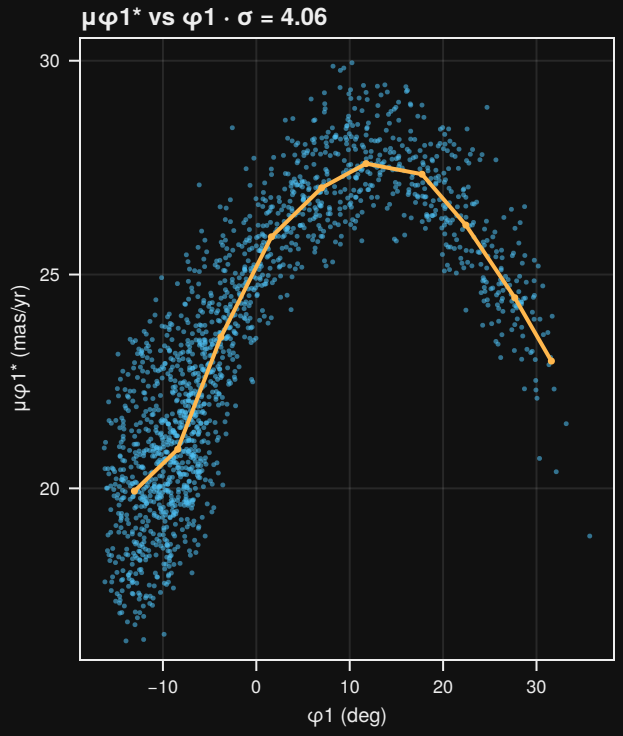
Stream frame (great-circle fit) · length 43.3° · width σ 3.39°

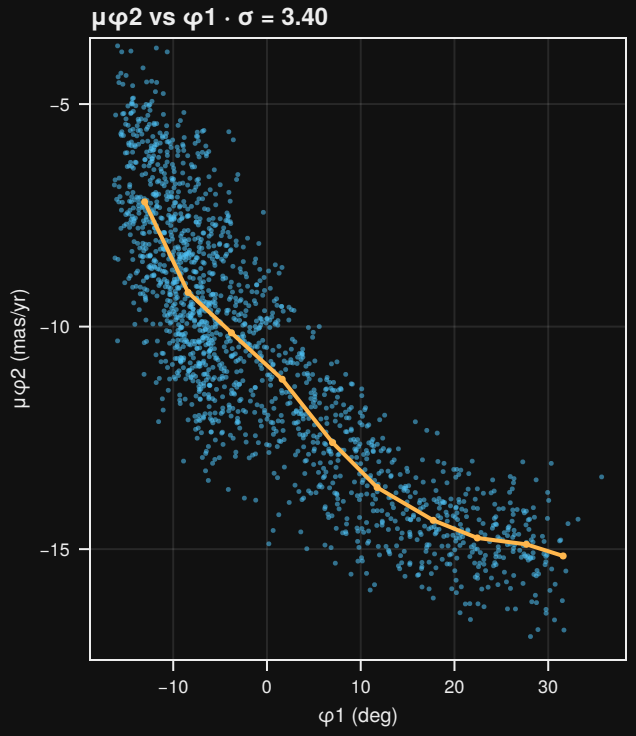


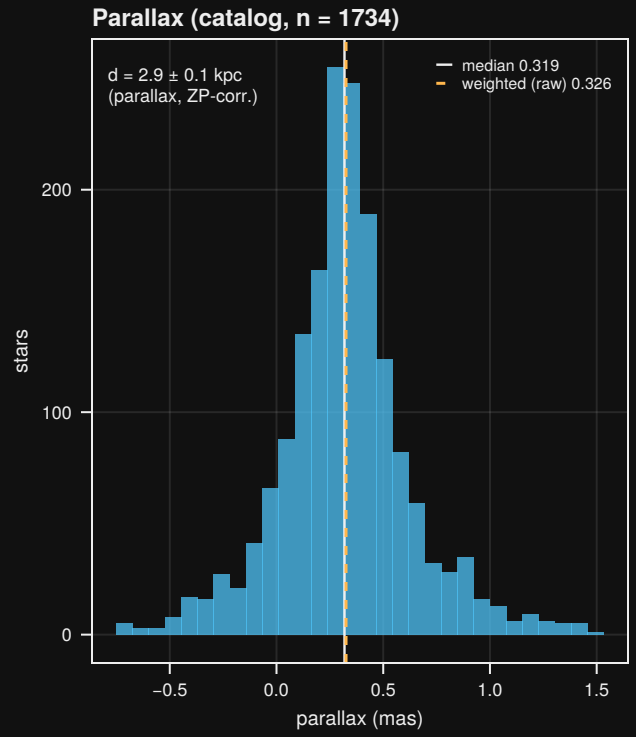
Gaia-9

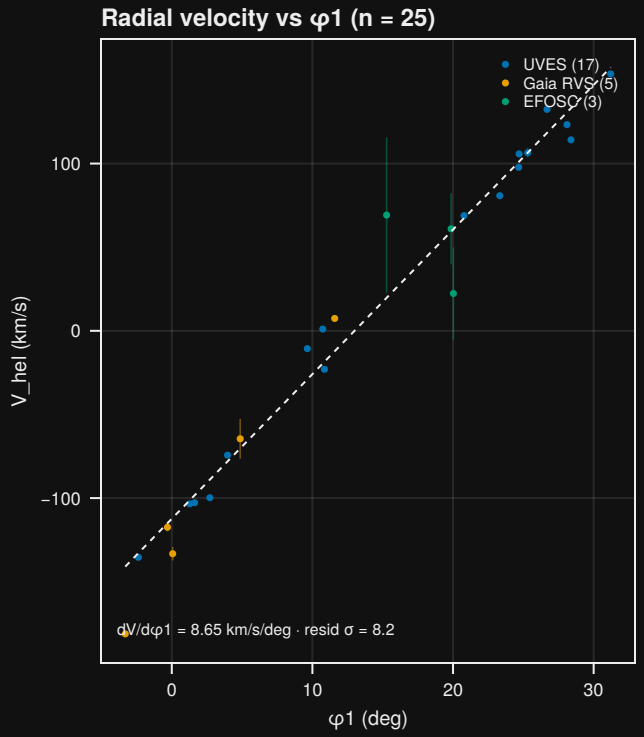
Kinematics · proper motion · parallax · radial velocity

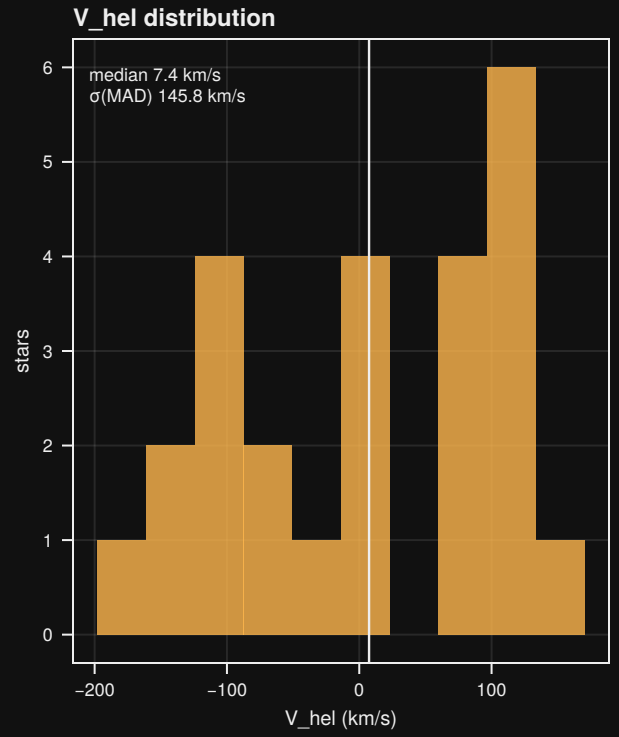






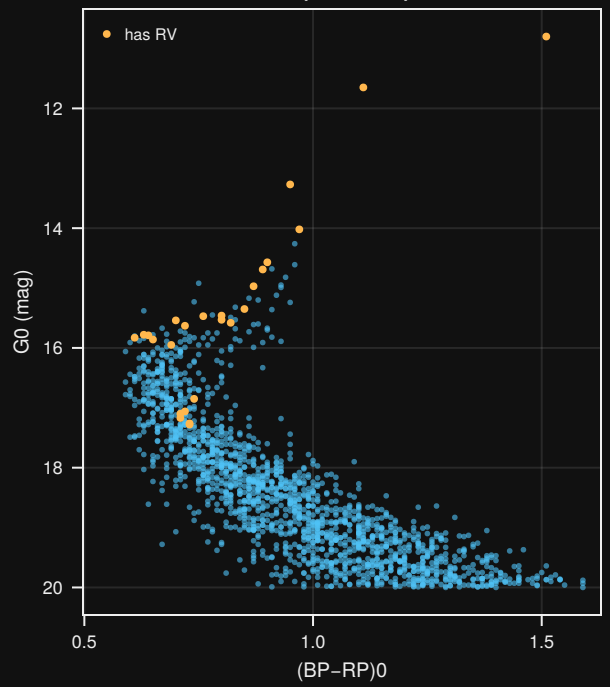




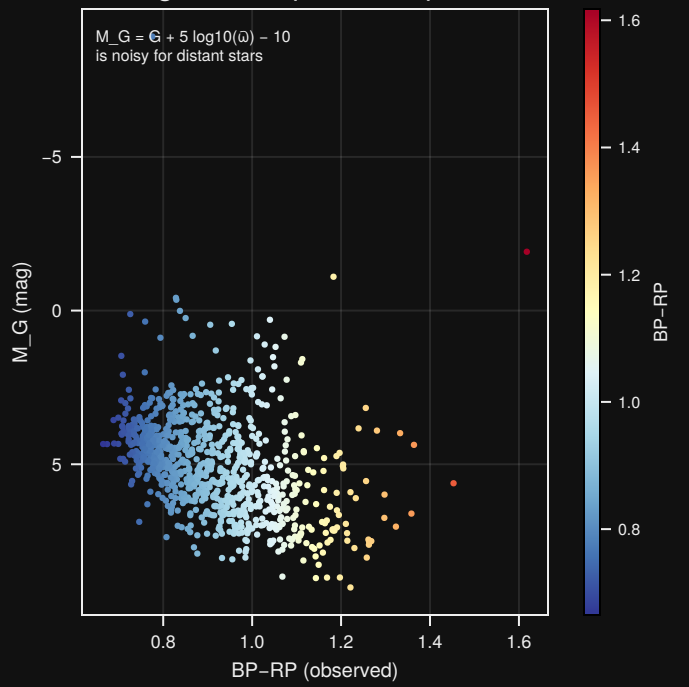


Gaia-9

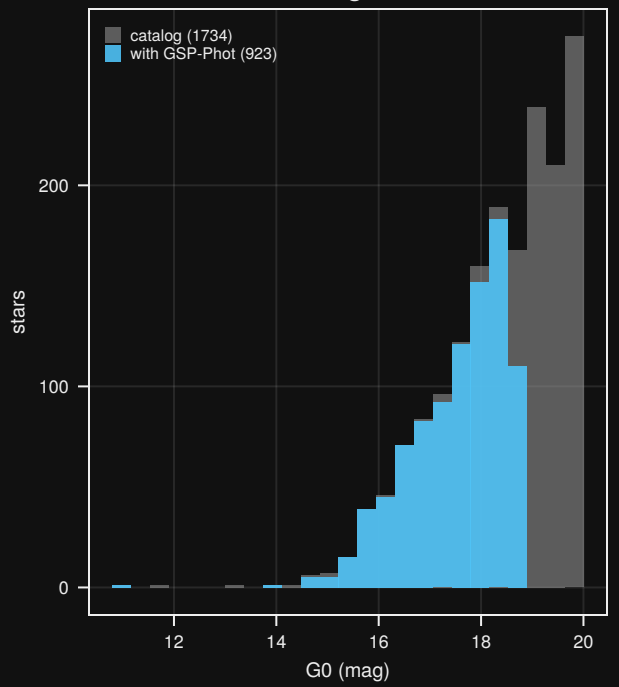
CMD · dereddened (n = 1734)



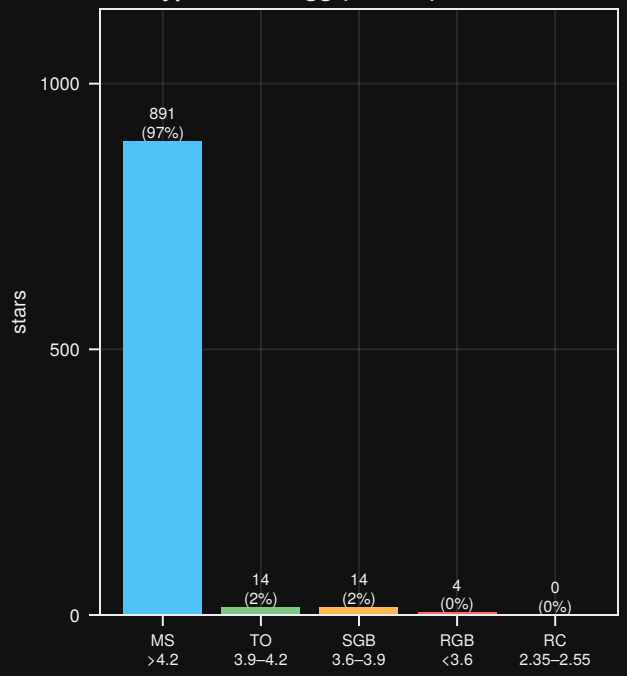
HR diagram · ϖ>0 (n = 894/923)



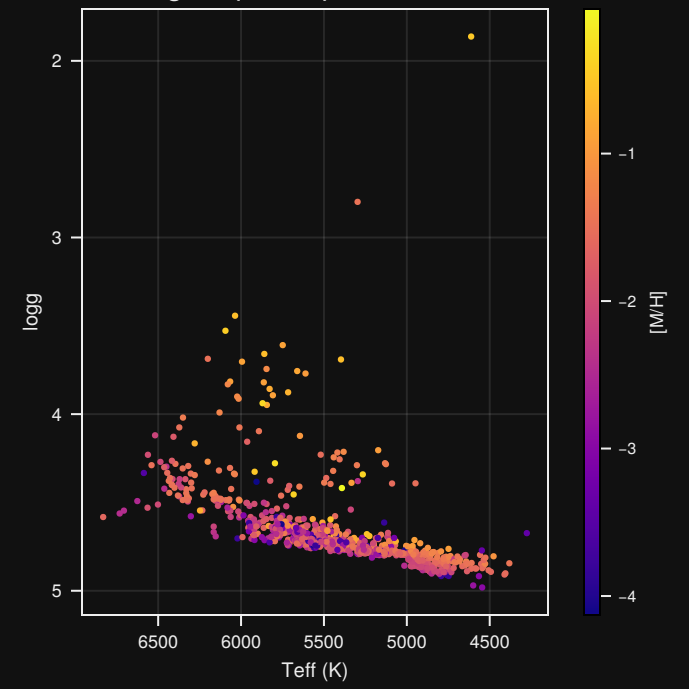
G0 & GSP-Phot coverage



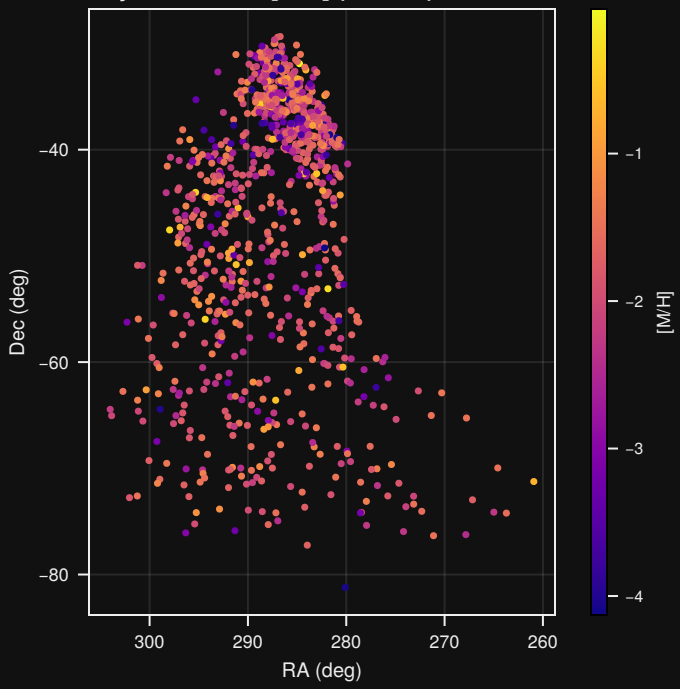
Star types from logg (n = 923)



Kiel diagram (n = 923)



Sky · GSP-Phot [M/H] (n = 923)

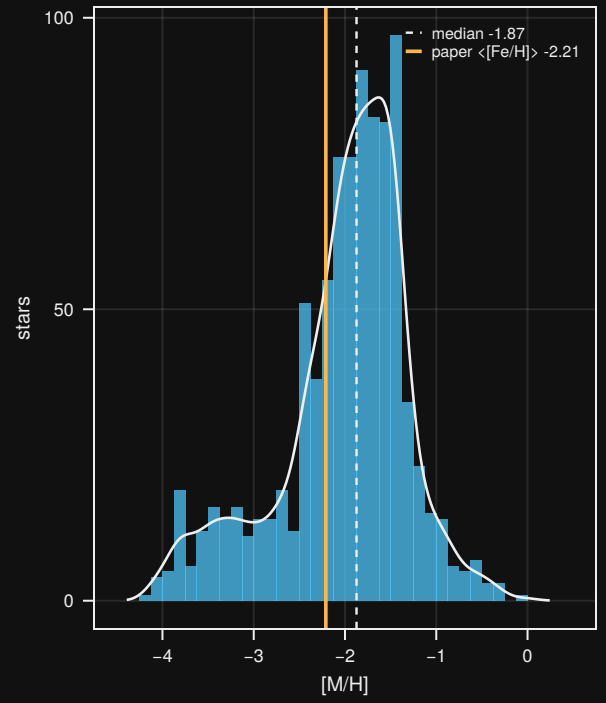


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

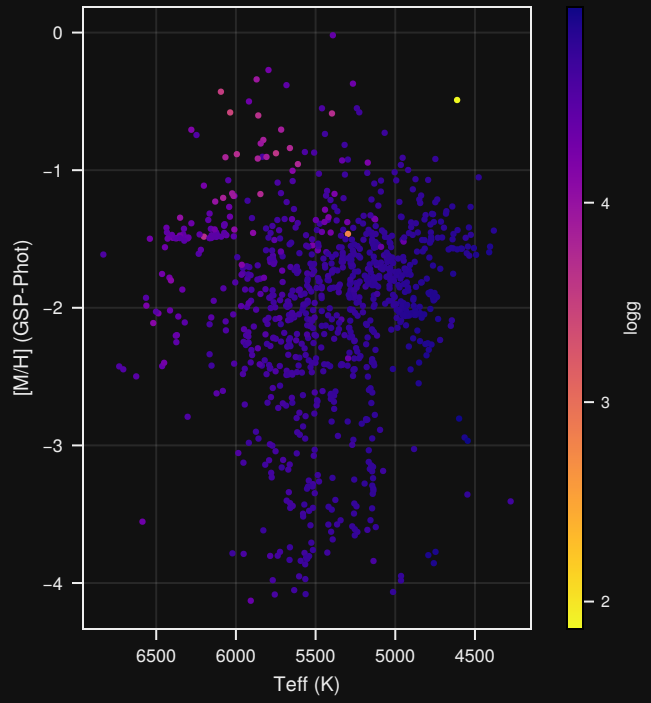
Gaia-9

Metallicity · age · astrometric quality

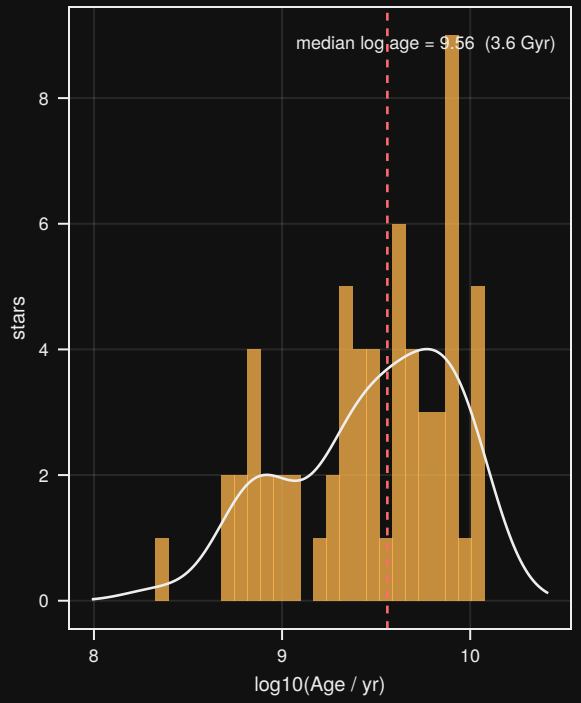
[M/H] GSP-Phot (n = 923)



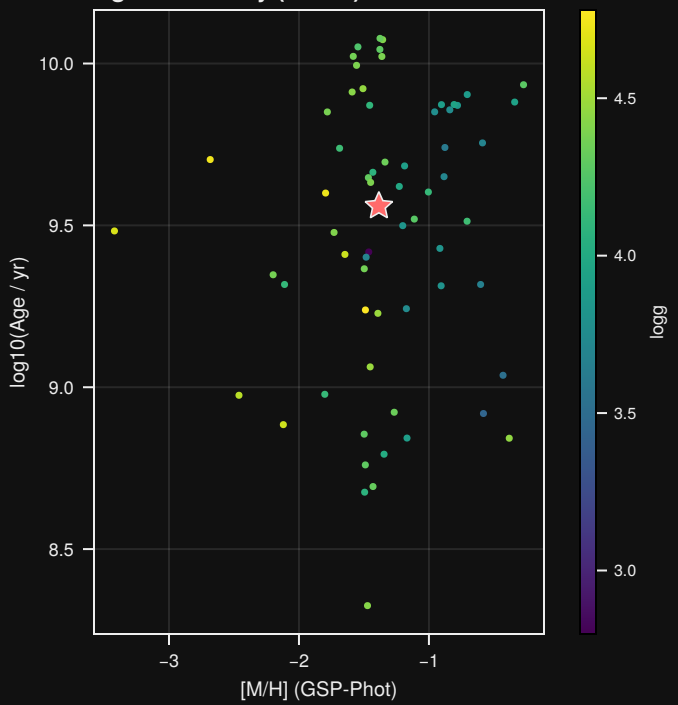
Teff vs [M/H] (n = 923)



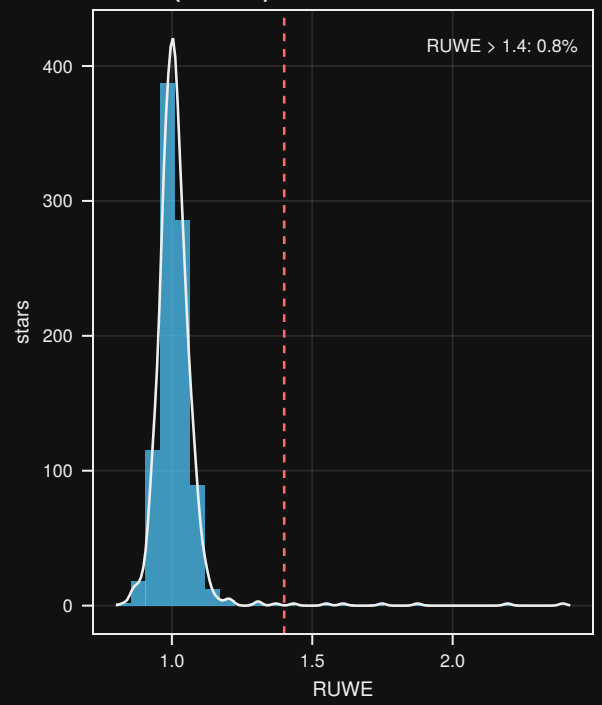
FLAME age (n = 64)



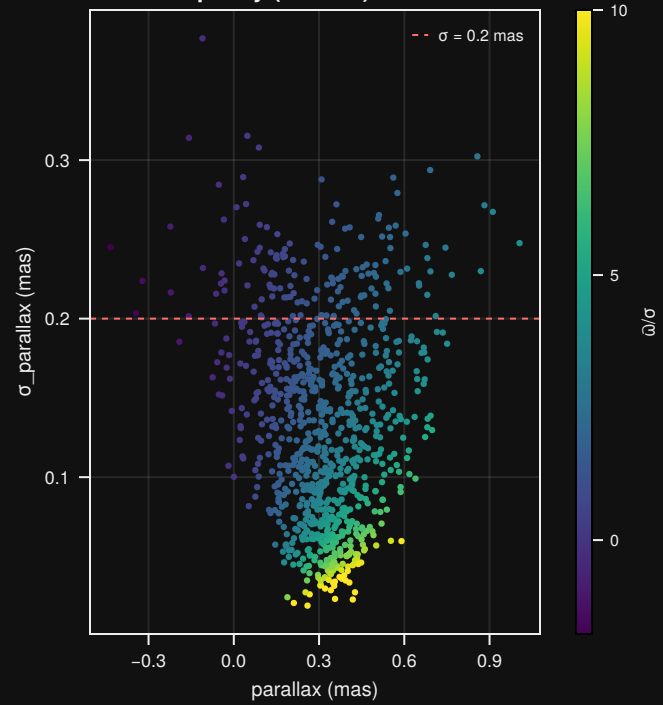
Age–metallicity (n = 64)



RUWE (n = 923)

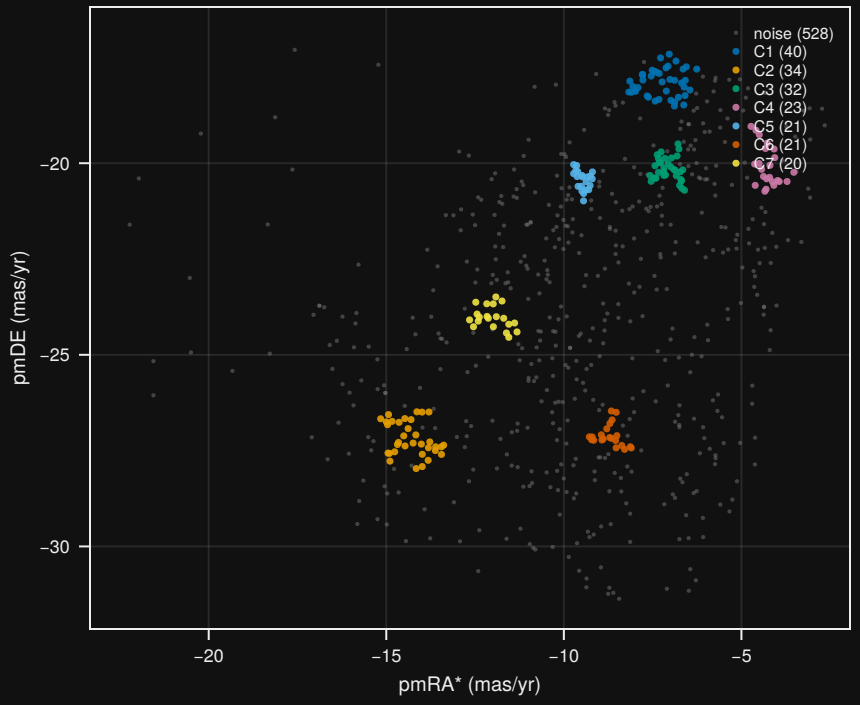


Parallax quality (n = 923)

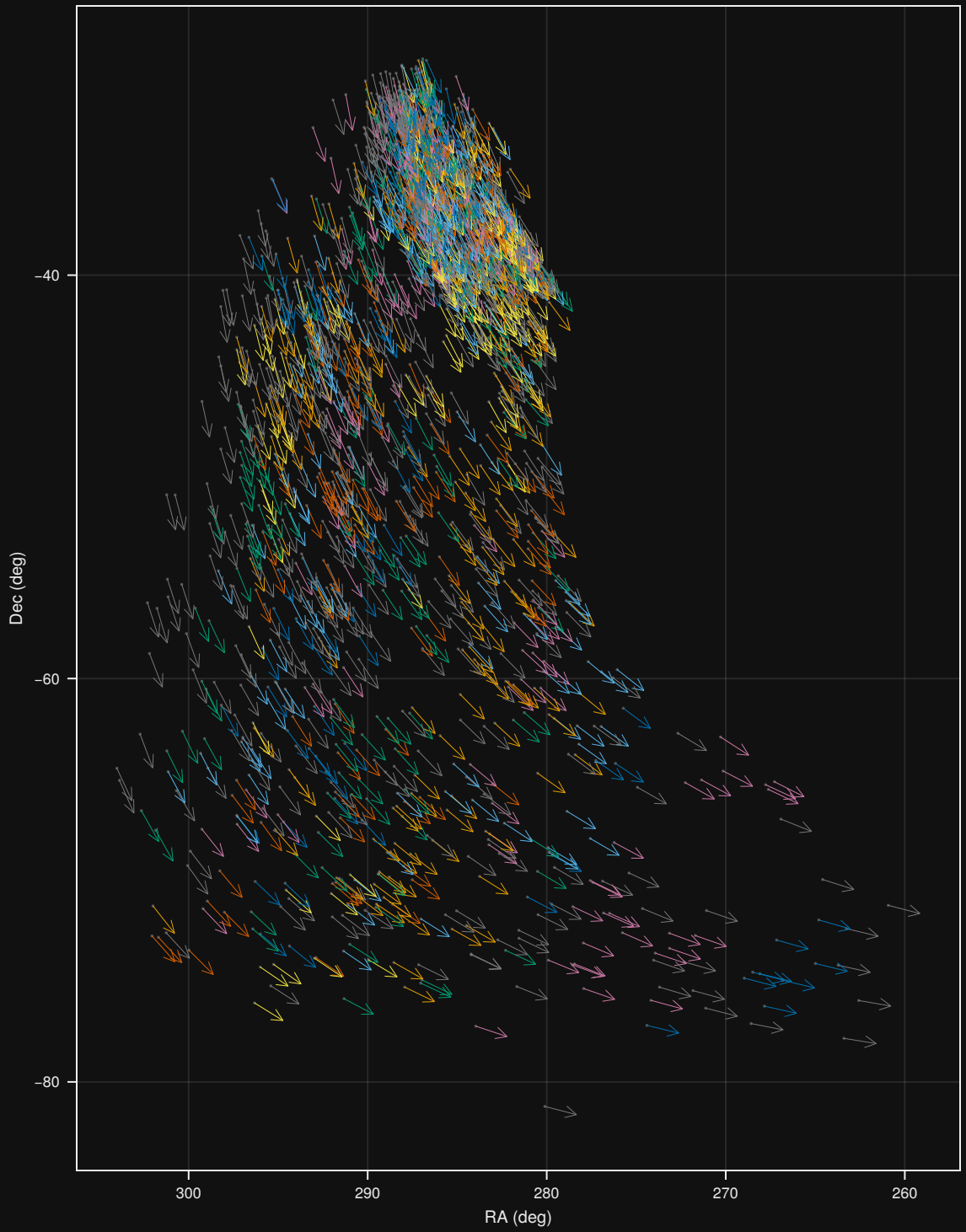


Gaia-9

PM clustering · HDBSCAN · 120 cluster(s) · noise 30%



Proper-motion directions (arrows) coloured by cluster



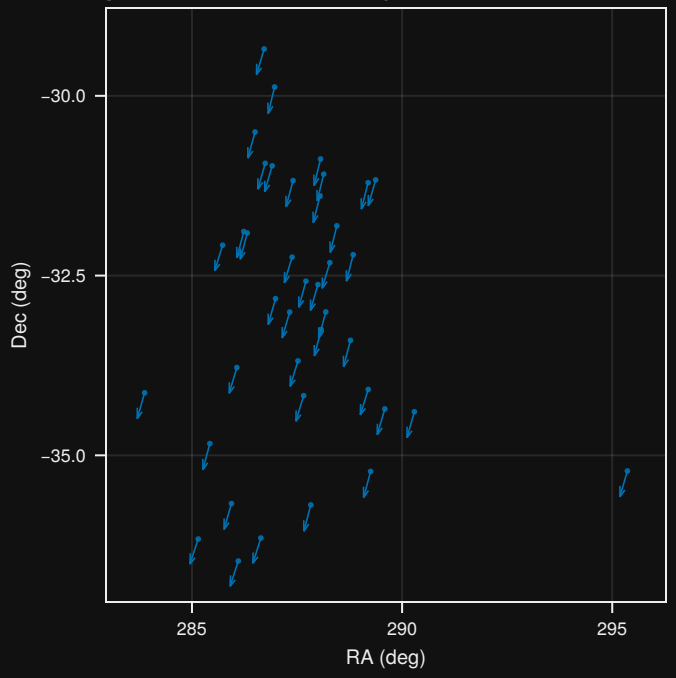
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	528	-9.39	-23.78	3.72	-1.85	9
C1	40	-7.18	-17.87	0.63	-2.16	0
C2	34	-14.27	-27.32	0.68	-1.97	1
C3	32	-7.10	-20.17	0.40	-1.83	0
C4	23	-4.29	-20.24	0.31	-1.60	0
C5	21	-9.48	-20.36	0.21	-2.25	0
C6	21	-8.67	-27.18	0.34	-1.85	1
C7	20	-11.99	-24.05	0.57	-1.95	1
C8	20	-8.07	-20.71	0.42	-1.80	0

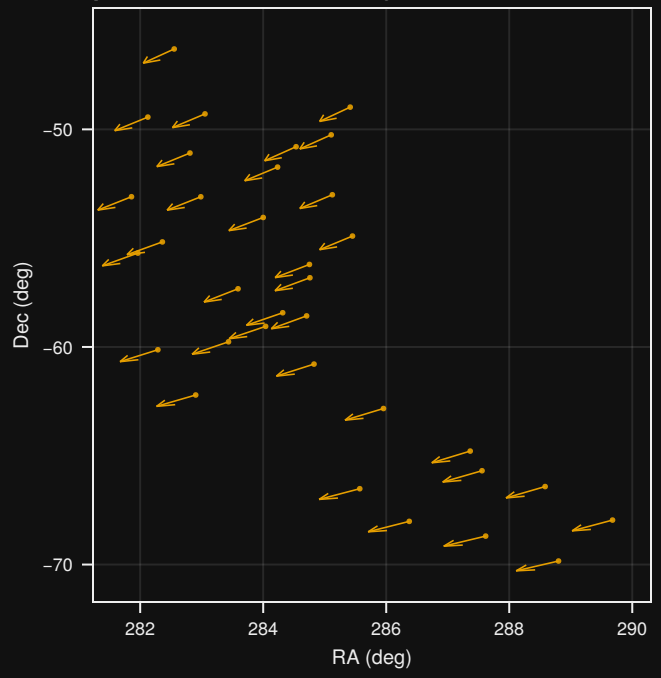
Gaia-9

Per-cluster proper-motion vector maps

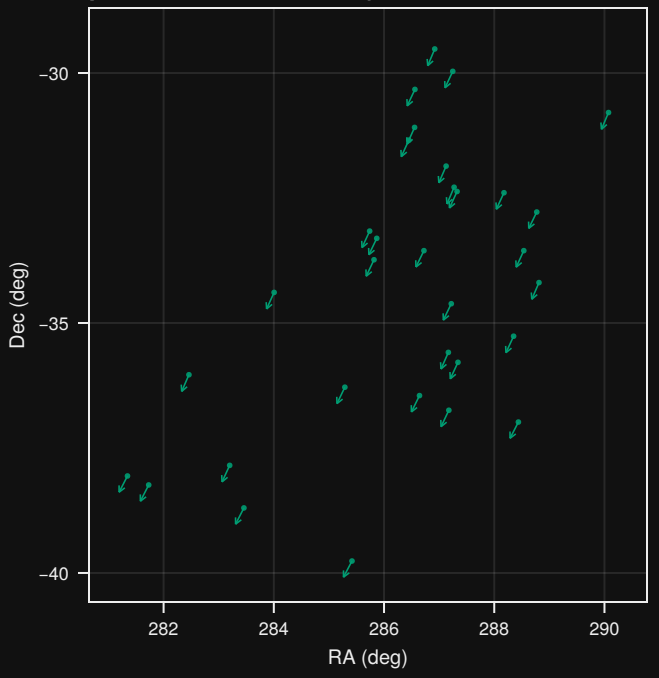
Cluster C1 · n = 40
pm = (-7.18, -17.87) mas/yr



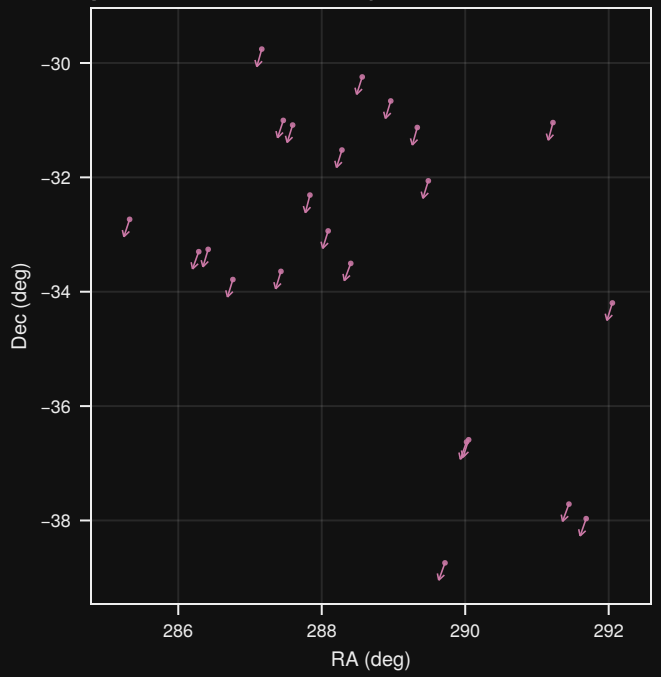
Cluster C2 · n = 34
pm = (-14.27, -27.32) mas/yr



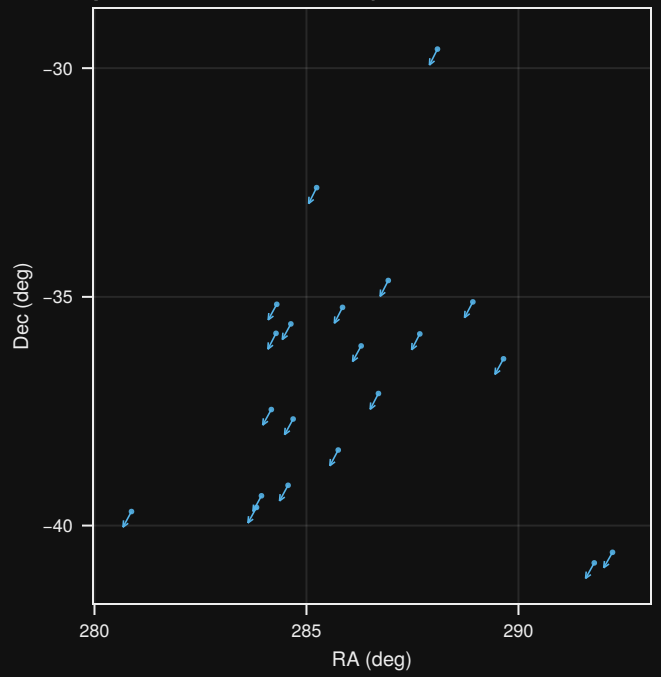
Cluster C3 · n = 32
pm = (-7.10, -20.17) mas/yr



Cluster C4 · n = 23
pm = (-4.29, -20.24) mas/yr



Cluster C5 · n = 21
pm = (-9.48, -20.36) mas/yr



Noise · n = 528

